Universitat Politècnica de València
Master's Degree in Fluid Mechanics

Aerodynamic Analysis of a Vehicle Overtaking Manoeuvre: A Comparative Study Between Quasi-Steady 3D and Transient 2D CFD Approaches

Group 12

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Abstract

The aerodynamic interaction between two vehicles during an overtaking manoeuvre is investigated by means of Computational Fluid Dynamics (CFD) simulations using the commercial solver STAR-CCM+. Two complementary modelling strategies are employed: a three-dimensional (3D) quasi-steady approach simulating five discrete longitudinal positions, and a two-dimensional (2D) transient simulation using the overset mesh technique. A Grid Convergence Index (GCI) study confirms discretisation uncertainty below 2.5 %. The 3D results identify three aerodynamic phases: an approach phase in which both vehicles exhibit isolated-like drag ($C_D \approx 0.35$); a lateral displacement phase in which bluff-body gap blockage drives both coefficients to a simultaneous peak at full lateral alignment ($C_{D,1} = 0.426$, $C_{D,2} = 0.433$, +23% relative to the approach); and a departure phase in which the drag distribution reverses as Car 2 assumes the leading position. The 2D transient simulation reproduces the canonical three-stage force signature — a 39 % drag reduction during slipstreaming, a 106 % spike at the transition, and progressive post-overtaking stabilisation — confirming its qualitative validity as a surrogate. A significant quantitative divergence in absolute C_D magnitudes (0.35–0.44 in 3D versus 1.87–2.08 in 2D) is attributed to the planar confinement constraint of the 2D model (piston effect), which prevents pressure relief over the roof and underbody. Cross-methodological comparison must therefore rely on normalised relative trends rather than absolute magnitudes.

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Nomenclature

Symbol	Definition	Units
C_D	Aerodynamic drag coefficient	–
C_L	Aerodynamic lift coefficient	–
F_D	Aerodynamic drag force	N
L	Vehicle length (reference length)	m
p	Apparent order of convergence	–
r	Grid refinement ratio	–
Re	Reynolds number	–
U_∞	Free-stream velocity	m/s
V_x	Longitudinal velocity component	m/s
V_{min}	Minimum longitudinal velocity in wake	m/s
Δx	Relative longitudinal position between vehicles	m

Abbreviations	
CFD	Computational Fluid Dynamics
GCI	Grid Convergence Index
MRF	Multiple Reference Frame
NRMSE	Normalised Root Mean Square Error
RANS	Reynolds-Averaged Navier–Stokes
SST	Shear Stress Transport

1 Introduction

The aerodynamic interaction between road vehicles in close proximity generates rapidly evolving forces that affect lateral stability, drag, and road safety [Hucho, 1998, Howell et al., 2012]. During an overtaking manoeuvre, the transient pressure field produces significant buffeting loads on both vehicles, with particular concern for lighter vehicles and large goods vehicles [Noger et al., 2005]. With the growing relevance of vehicle platooning in autonomous driving and freight transport, accurate prediction of inter-vehicle aerodynamic loads has become a pressing engineering challenge [Schito and Braghin, 2016, Browand et al., 2004].

A fully resolved 3D transient simulation of the overtaking problem is computationally prohibitive for routine studies [Watkins and Vino, 2008, Corin et al., 2008]. A key question is whether a quasi-steady approximation is valid: following Greitzer et al. [2007], the reduced frequency $\beta = (L/U_\infty)\,\omega_{\text{exc}} \approx 0.10$ places the problem precisely at the quasi-steady boundary, constituting the central scientific motivation of this work.

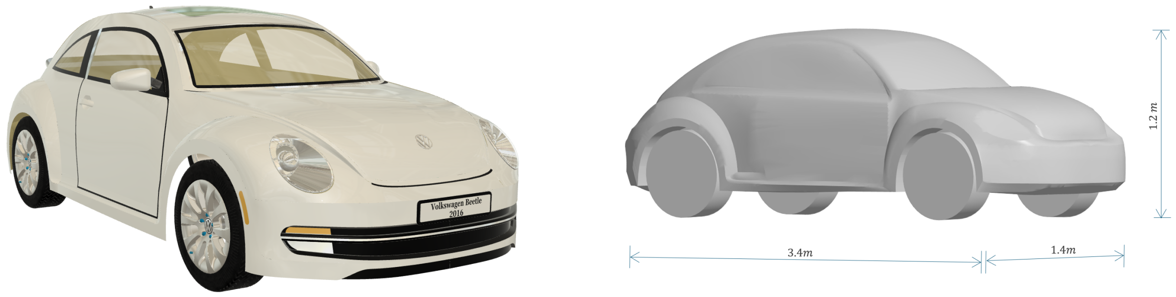
The objective is to assess whether a 2D transient overset simulation can serve as a reliable surrogate for a 3D quasi-steady parametric analysis of the overtaking manoeuvre. Five discrete positions are simulated with 3D steady-state RANS and the complete manoeuvre with a 2D transient overset model; drag coefficients and inter-vehicle velocity fields are then compared quantitatively. The study addresses four objectives: (i) characterise C_D vs. relative position using 3D RANS at five positions; (ii) simulate the full manoeuvre in 2D transient overset; (iii) assess mesh independence via GCI; and (iv) evaluate whether the 2D model is a reliable and computationally efficient surrogate.

The remainder of the report is structured as follows. Section 2 describes the baseline configuration. Section 3 presents the mesh independence study. Section 4 presents and compares results of both approaches. Section 6 summarises the conclusions.

2 Baseline Case Configuration

2.1 Vehicle Geometry

The geometry is based on a Volkswagen Beetle 2016 ($L = 3.4\,\text{m}$, $W = 1.4\,\text{m}$, $H = 1.2\,\text{m}$) sourced from GrabCAD [GrabCAD, 2024] and defeatured following standard practice: mirrors, panel gaps, and tyre detail removed; tyres replaced by smooth cylinders. This preserves the macroscopic bluff-body wake while reducing cell count (Figure 1).



a) Original CAD

b) Simplified geometry

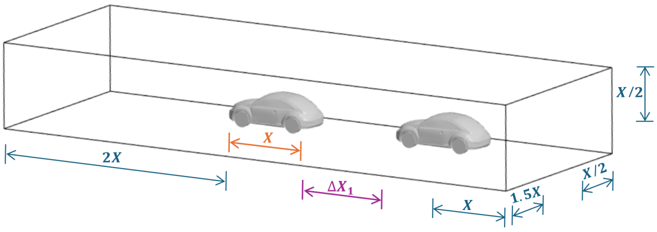
Figure 1: Original and defeatured Volkswagen Beetle 2016.

2.2 Computational Domain

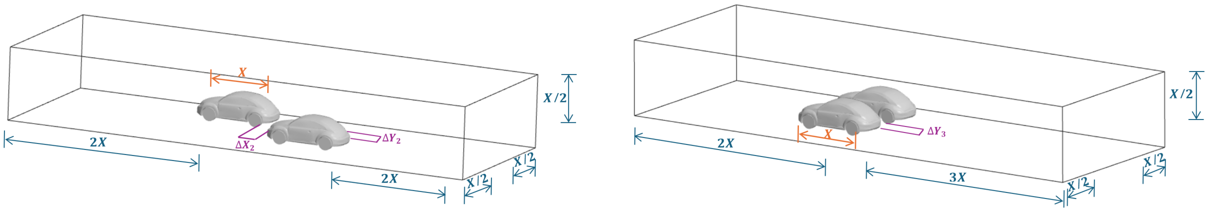
The domain measures $20\,\mathrm{m} \times 7\,\mathrm{m} \times 3\,\mathrm{m}$ ($5.9L \times 2.1L \times 0.9L$), placing the inlet $2L$ upstream of the leading vehicle. The five positions are shown in Figure 2 and their inter-vehicle separations in Table 1. The combined frontal area of both vehicles ($3.36\,\mathrm{m}^2$) against the domain cross-section ($21\,\mathrm{m}^2$) yields BR $\approx 16\%$, exceeding the recommended 5% [Hucho, 1998, Blocken and Gualtieri, 2012]; this is acknowledged as a limitation, partially mitigated by slip-wall conditions on lateral and top faces.

Table 1: Surface-to-surface separations for the five positions ($L = 3.4\,\mathrm{m}$).

Case	ΔX **[m]**	$\Delta X/L$	ΔY **[m]**	$\Delta Y/L$	**Description**
P1	3.40	1.00	0.00	0.00	Approach, same lane
P2	0.78	0.23	1.00	0.30	Lateral displacement initiated
P3	0.00	0.00	1.00	0.30	Full lateral alignment
P4	0.03	0.01	1.00	0.30	Car 2 nose just ahead
P5	1.27	0.37	0.00	0.00	Departure, same lane

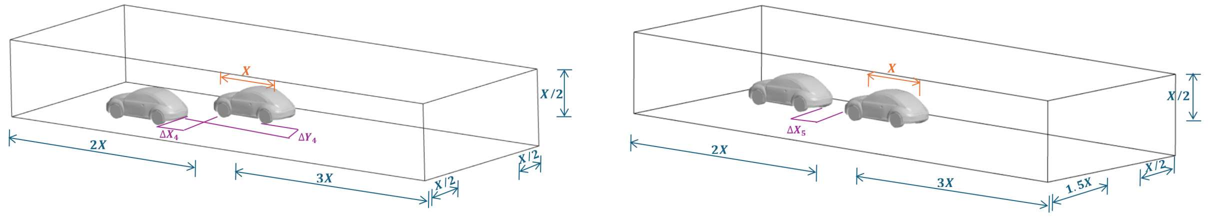

X/2
X
2X
ΔX_1
X
1.5X
X/2

(a) P1


X
X/2
ΔX_2
ΔY_2
2X
2X
X/2
X/2
X/2
ΔY_3
X
2X
3X
X/2
X/2

(b) P2

(c) P3


X
X/2
ΔX_4
ΔY_4
2X
3X
X/2
X/2
X
X/2
ΔX_5
2X
3X
1.5X
X/2

(d) P4

(e) P5

Figure 2: Computational domain for the five quasi-steady positions. $X = L = 3.4\,\mathrm{m}$; purple arrows ΔX_i, ΔY_i as in Table 1.

2.3 Boundary Conditions

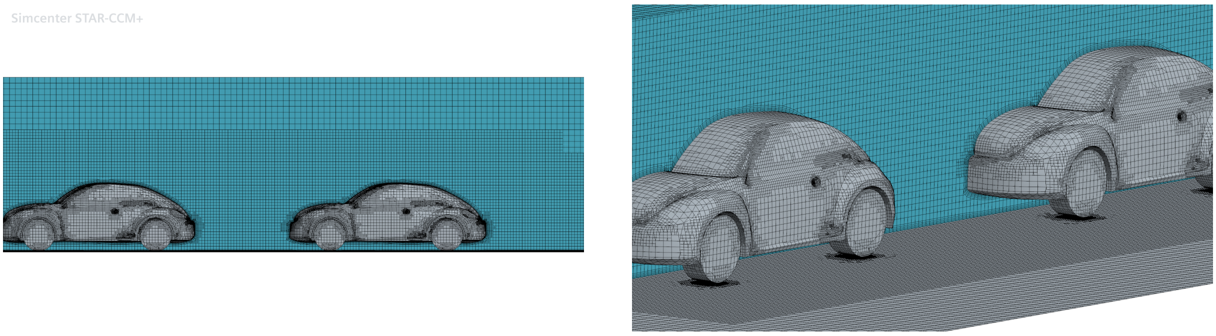
The reference frame is fixed to Car 2 [Schütz, 2015]. Table 2 summarises all boundary conditions. The velocity inlet imposes $U_\infty = 33.33\,\mathrm{m/s}$ (120 km/h); the moving ground translates at the same speed to suppress a spurious ground boundary layer [Katz, 2006]. Car 1 surfaces carry a tangential velocity of $+5.55\,\mathrm{m/s}$ (20 km/h) to reproduce the correct skin-friction field in the relative frame.

Table 2: Boundary conditions.

Boundary	**Type**	**Value**
Inlet	Velocity inlet	$33.33\,\mathrm{m/s}$
Outlet	Pressure outlet	0 Pa (gauge)
Ground	Moving no-slip wall	$33.33\,\mathrm{m/s}$
Car 2	No-slip wall	0 m/s
Car 1	No-slip wall	$U_{\tan} = +5.55\,\mathrm{m/s}$
Lateral & top	Slip wall	$\tau = 0$, $U_n = 0$

2.4 Mesh Strategy

A trimmed hexahedral mesh was generated in STAR-CCM+ (Trimmed Cell Mesher, Surface Remesher, Prism Layer Mesher). Global base size $\delta_{\text{base}} = 0.1\,\text{m}$; three prism layers on vehicle surfaces and ground with stretching ratio 1.2, targeting $30 \lesssim y^+ \lesssim 300$ consistent with the wall-function approach. Two volumetric refinement levels: outer bounding boxes at 70 mm and inner boxes plus Car 1 wake region at 50 mm (Figure 3).



(a) Symmetry-plane overview.

(b) Surface mesh detail.

Figure 3: Medium mesh (825 015 cells). Base 0.1 m; outer 70 mm; inner/wake 50 mm; 3 prism layers, ratio 1.2.

2.5 Turbulence Model and Numerical Settings

The Realizable k-ε model [Shih et al., 1995] was adopted. The k-ω SST model [Menter, 1994] failed to converge due to insufficient near-wall resolution at the wheel–ground clearance and sensitivity of the ω equation to free-stream conditions (residual histories in Annex A); this introduces a known accuracy penalty of ∼21% vs. ∼9% for SST [Abdul-Rahman et al., 2021], acknowledged as a limitation. Flow was treated as steady, incompressible, and isothermal (Mach < 0.3); second-order upwind convective scheme; SIMPLE pressure–velocity coupling. Convergence criterion: residuals $< 10^{-4}$ and asymptotic stabilisation of C_D [Celik et al., 2008].

3 Mesh Independence Study

3.1 Methodology

A dual-criteria mesh independence study [Oberkampf and Roy, 2010] was conducted on position P1: global convergence via C_D (GCI, Celik et al. 2008) and local convergence via the inter-vehicle V_x profile (NRMSE). Three meshes with refinement ratio $r \approx \sqrt{2}$ and safety factor $F_s = 1.25$: Coarse (457 443 cells), Medium (825 015), Fine (3.6M).

3.2 Results

Table 3: C_D across mesh levels and pointwise GCI on V_{min}.

		Coarse	**Medium**	**Fine**	**Med–Fine dev.**
C_D	Car 1	0.317	0.295	0.293	0.85%
	Car 2	0.364	0.361	0.361	<0.1%
V_{min} [m/s]		21.98	20.44	21.01	$GCI_{21} = 2.04\%$

Car 1 shows non-monotonic convergence (Coarse departs 8.4% from Fine); a formal three-grid GCI is therefore not applicable [Celik et al., 2008]. The Medium–Fine pair confirms stabilisation at 0.85%, within the 5% threshold. Car 2 is converged at Coarse level ($< 0.9\%$ variation). The V_x NRMSE between Coarse and Fine is $\approx 4\%$ ($< 5\%$ threshold, Oberkampf and Roy 2010); the pointwise $GCI_{21} = 2.04\%$ with asymptotic ratio 1.03 confirms adequate local resolution (Figure 4).

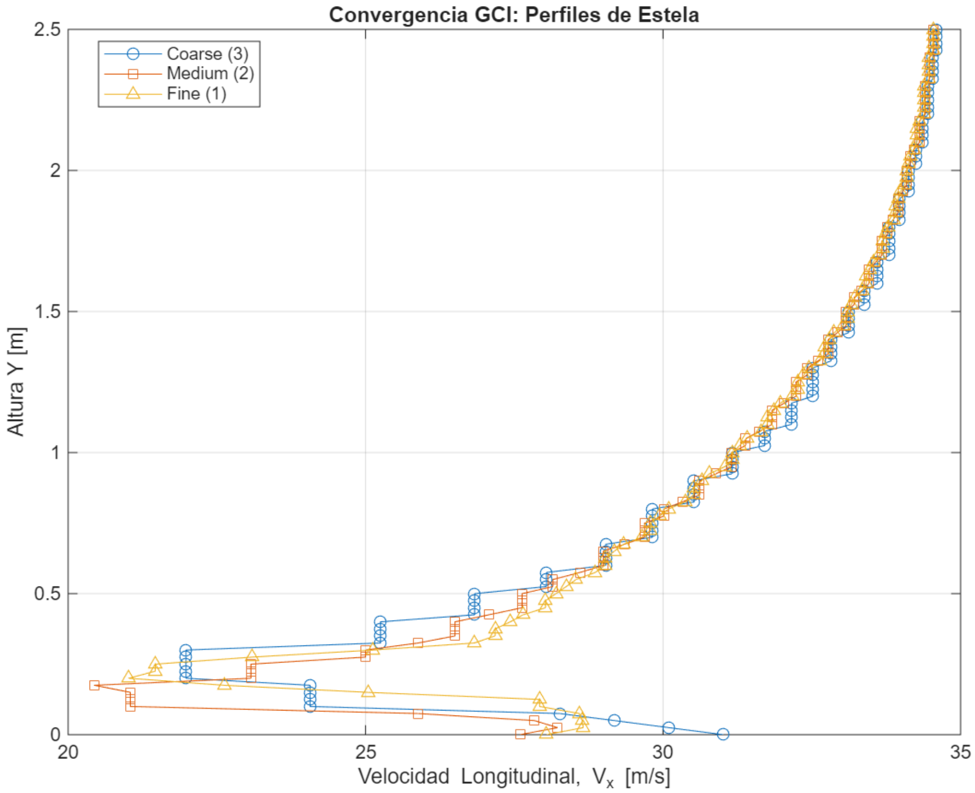


Figure 4: Inter-vehicle V_x profiles for the three mesh levels.

3.3 Selected Mesh

The **Coarse mesh** (457 443 cells, 5.3 CPU-h vs. 16.0 CPU-h for Fine, 16 cores) is selected for all parametric simulations. Its $\approx 8\%$ absolute error on Car 1 C_D is acknowledged;

since all positions share the same mesh, the *relative* C_D trends across positions — the primary output of this study — are considered reliably captured.

4 Parametric Study: Quasi-Steady 3D Positions

4.1 Evolution of the Drag Coefficient with Overtaking Position

The converged drag coefficients are presented in Figure 5 and Table 4. Three aerodynamic phases are identified.

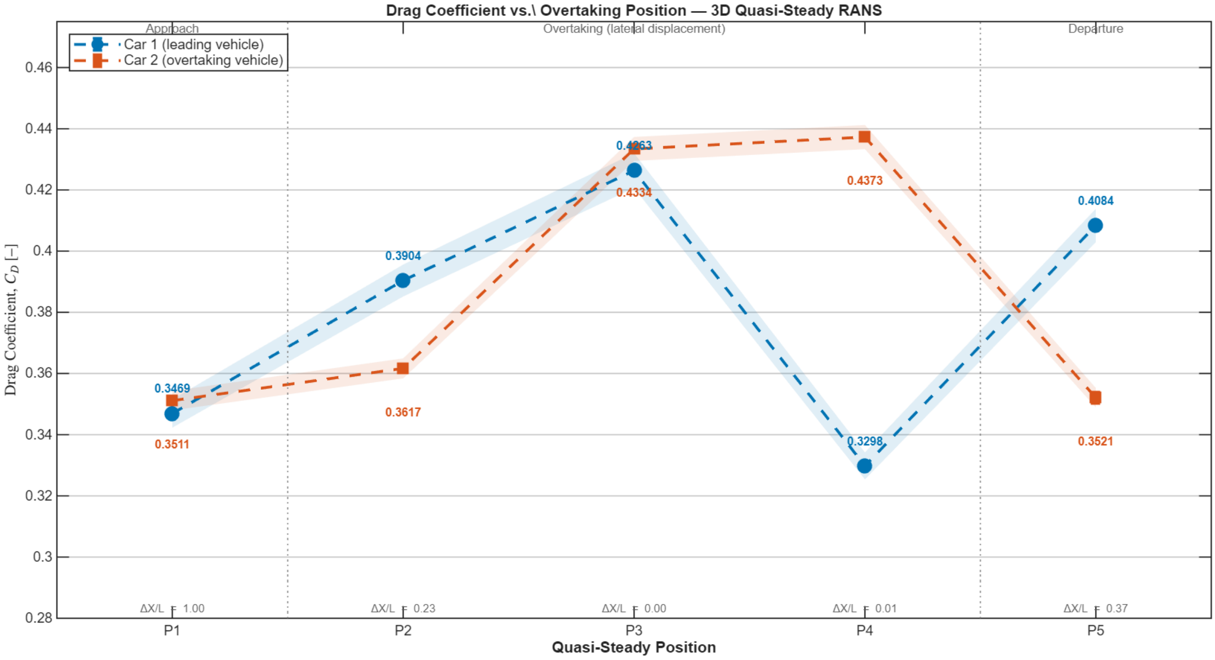


Figure 5: C_D of Car 1 (leading) and Car 2 (overtaking) vs. quasi-steady position. Error bars indicate iterative standard deviation σ from Table 4. Dashed lines connect discrete points and do not imply a continuous solution.

Table 4: Converged C_D (mean of last 50 iterations) and iterative standard deviation σ for both vehicles.

Case	$\Delta X/L$	$\Delta Y/L$	C_D **Car 1**	σ_1	C_D **Car 2**	σ_2
P1	1.00	0.00	0.347	0.00081	0.351	0.00085
P2	0.23	0.30	0.390	0.00007	0.362	0.00012
P3	0.00	0.30	0.426	0.00027	0.433	0.00010
P4	0.01	0.30	0.330	0.00002	0.437	0.00001
P5	0.37	0.00	0.408	0.00057	0.352	0.00176

Phase I — Approach (P1). Both vehicles travel in the same lane with Car 2 one vehicle-length behind. C_D values are nearly equal and close to the isolated-vehicle reference, indicating only partial slipstreaming attenuation at $\Delta X/L = 1.00$ [Browand et al., 2004].

Phase II — Lateral displacement (P2–P4). The inter-vehicle gap restricts the transverse expansion of Car 2's bow wave (*bluff-body blockage effect* [Hucho, 1998, Corin et al., 2008]): the stagnation pressure at Car 2's nose rises, increasing $C_{D,2}$. Simultaneously, this high-pressure region acts on Car 1's base, reducing $C_{D,1}$ via a propulsive pressure surge [Watkins and Vino, 2008, Noger et al., 2005]. Both effects peak at P3 (full lateral alignment): $C_{D,2} = 0.433$ (+23% relative to P1) and $C_{D,1} = 0.426$ (+23%). At P4 the interaction reverses: $C_{D,1}$ drops sharply to 0.330 while $C_{D,2}$ reaches its absolute maximum of 0.437.

Phase III — Departure (P5). Car 2 leads and Car 1 enters its wake. The drag distribution reverses with respect to P1: $C_{D,1} = 0.408$ (trailing, partial slipstream benefit) and $C_{D,2} = 0.352$ (leading, free-stream exposure). The residual elevation of $C_{D,1}$ indicates $\Delta X/L = 0.37$ is insufficient for full wake recovery [Browand et al., 2004].

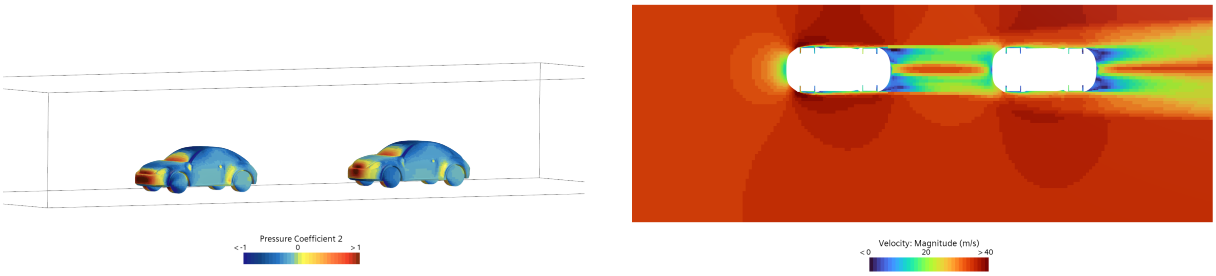
4.2 Flow Field Analysis

Figure 6 presents the surface pressure coefficient (C_p) and top-view velocity magnitude at positions P1, P3, and P5.

P1. Symmetric stagnation lobes ($C_p \approx +1$) at each front face; Car 2 fully immersed in Car 1's momentum deficit, consistent with $C_{D,1} = C_{D,2} \approx 0.35$. Notably, the top-view velocity field reveals a narrow, rapidly closing wake behind each vehicle — markedly shorter than the extended vortical structures observed in the 2D overset simulation (Section 5.4). This is physically expected: the 3D top-view slice at mid-height captures only the in-plane momentum deficit, while the bulk of the wake energy is redistributed vertically through roof and underbody flow paths unavailable in the 2D planar model. The apparent wake brevity is therefore a visualisation artefact of the cutting plane rather than a sign of insufficient domain length.

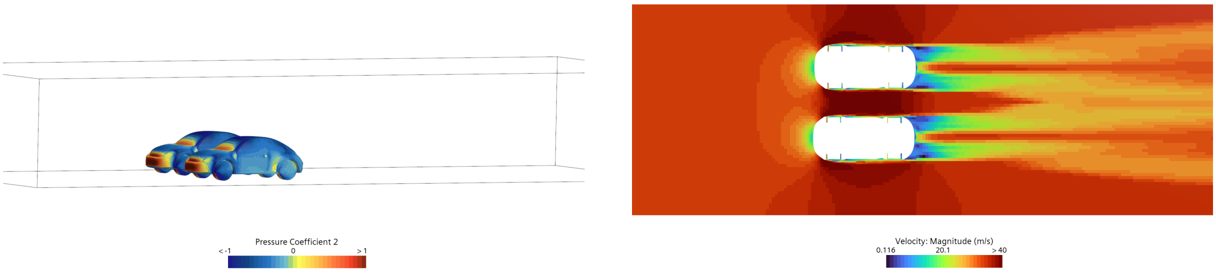
P3. The inter-vehicle gap acts as a convergent channel (Venturi effect), generating an intense low-pressure corridor on the inboard flanks and driving both C_D values to their peak. The pronounced inboard–outboard C_p asymmetry is the origin of the lateral attractive force documented in Noger et al. [2005], Watkins and Vino [2008].

P5. Car 2 recovers toward an isolated-vehicle C_p distribution; Car 1 shows a weakened front stagnation lobe consistent with partial wake shielding. Residual $C_{D,1}$ elevation confirms $\Delta X/L = 0.37$ is insufficient for full wake recovery [Browand et al., 2004]. The narrow top-view wake signature persists across all positions, underscoring the fundamental topological difference between the 3D and 2D flow fields and reinforcing the conclusion that absolute C_D magnitudes cannot be directly compared between the two approaches.


Pressure Coefficient 2
< -1
0
> 1
Velocity: Magnitude (m/s)
< 0
20
> 40

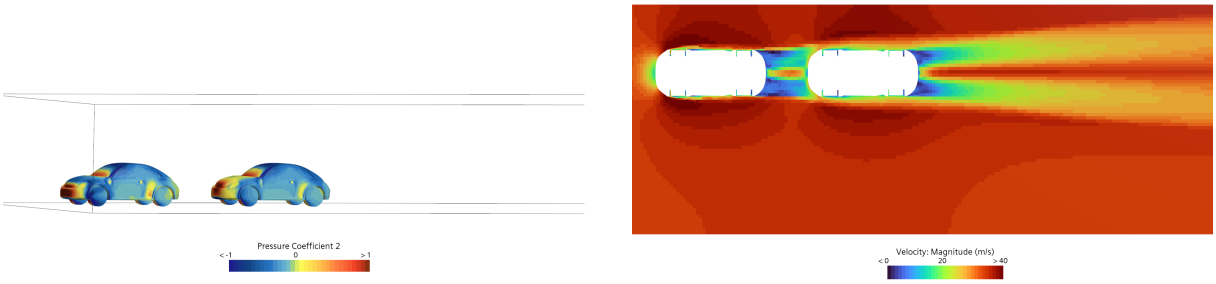
P1 — C_p surface

P1 — Velocity magnitude (top view)


Pressure Coefficient 2
< -1
0
> 1
Velocity: Magnitude (m/s)
0.116
20.1
> 40

P3 — C_p surface

P3 — Velocity magnitude (top view)


Pressure Coefficient 2
< -1
0
> 1
Velocity: Magnitude (m/s)
< 0
20
> 40

P5 — C_p surface

P5 — Velocity magnitude (top view)

Figure 6: Surface C_p (left) and top-view velocity magnitude (right) for P1, P3, and P5. Flow: left to right. $U_\infty = 33.33\,\text{m/s}$.

5 Transient 2D Overset Simulation

5.1 Modelling Approach, Reference Frame, and Solver Settings

A 2D transient simulation was conducted using the overset mesh in Simcenter STAR-CCM+. Each vehicle is modelled as a rectangular bluff body in top-view cross-section, neglecting underbody flow, roof vortices, and wheel-arch interactions — justified as a first-order approximation of the dominant in-plane pressure interaction [Corin et al., 2008]. A Galilean frame transformation keeps the domain compact: the inlet was set to $U_1 = 27.78$ m/s (Car 1 speed) and Car 2 prescribed a rigid-body translation at $\Delta U = 5.56$ m/s. The Realizable k-ε model [Shih et al., 1995] was adopted for consistency with the 3D cases. Boundary conditions are given in Table 5.

Table 5: Boundary conditions for the 2D transient overset simulation.

Boundary	**Type**	**Value / Setting**
Inlet	Velocity inlet	$U_\infty = 27.78$ m/s
Outlet	Pressure outlet	$p = 0$ Pa (gauge)
Top & Bottom	Slip wall	$\partial U/\partial n = 0,\ V_n = 0$
Vehicle surfaces	No-slip wall	$U = 0$ m/s
Overset interfaces	Overset Mesh	Interpolated (distance-weighted)

5.2 Mesh Strategy

The trimmed-cell mesher was used for the background domain and both overset regions. Three mesh levels: (i) coarse background at $\Delta x_{\text{bg}} = 0.20$ m; (ii) a Volumetric Control strip along the overtaking corridor refined to $\Delta x_{\text{vc}} = 0.07$ m, enforcing the 1:1 cell-size ratio required for accurate interpolation [Steger and Dougherty, 1983]; and (iii) the overset regions themselves, refining to $\Delta x_{\text{surf}} = 0.01$ m at the vehicle contour. A 5-layer prism mesh ($\delta_{\text{prism}} = 0.03$ m, ratio 1.3) targets $30 \leq y^+ \leq 300$. Mesh resolution was calibrated to match the 3D Coarse mesh in the inter-vehicle region (Figure 7).

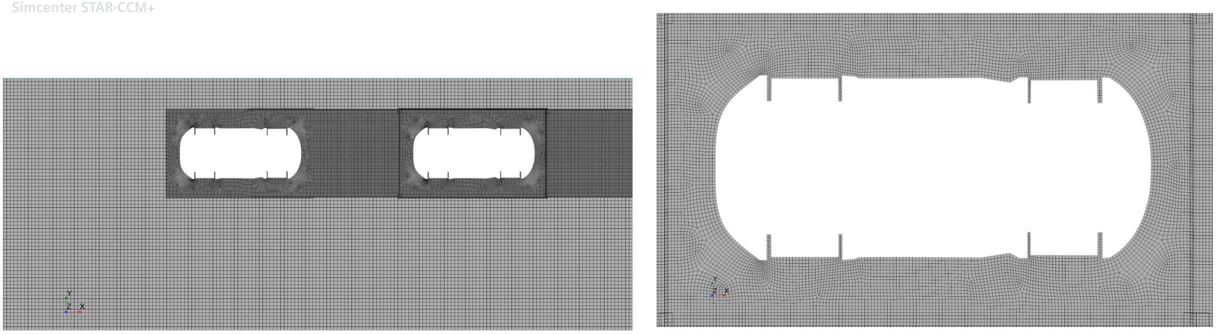


Figure 7: Left: global mesh with overset regions and Volumetric Control strip (cell size 0.07 m). Right: zoomed view showing the prism-layer-to-overset-to-background cell transition.

5.3 Initialisation and Time-Step Selection

A steady-state solution at P1 was used as initial condition [Corin et al., 2008] (Figure 14, Annex A). The Implicit Unsteady solver was then activated with $\Delta t = 1\times10^{-4}$ s, targeting Co ≈ 1 in the inter-vehicle gap. Peak values $\text{Co}_{\max} \approx 12$ are confined to two or three prism-layer cells at the sharp rear corners (geometric singularities); inner-loop residuals remained below 10^{-4} per time step throughout [Celik et al., 2008].

5.4 Flow Field Evolution

Figure 8 shows two key snapshots of the static pressure field. At $t = 0$ s Car 2 is sheltered in the wake (drafting mechanism). By $t = 0.22$ s the inter-vehicle gap develops an intense low-pressure corridor ($p \approx -4730$ Pa) driven by Venturi acceleration, generating a lateral attractive force on both vehicles — the primary aerodynamic destabilisation mechanism during overtaking [Noger et al., 2005, Watkins and Vino, 2008].

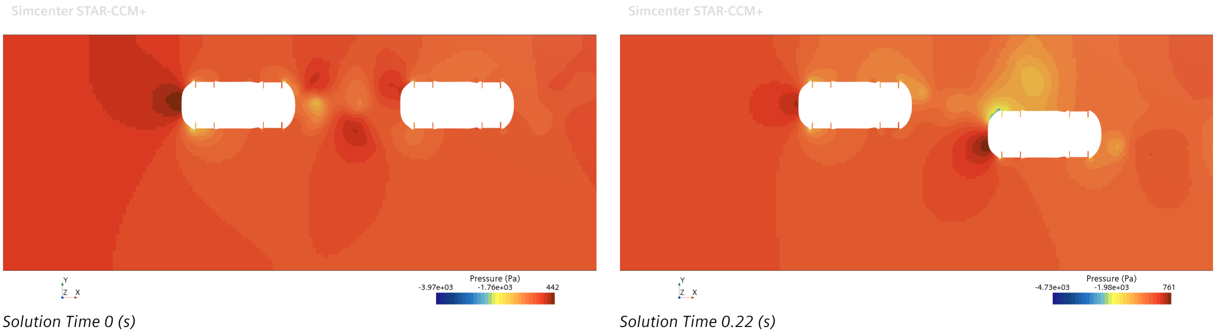
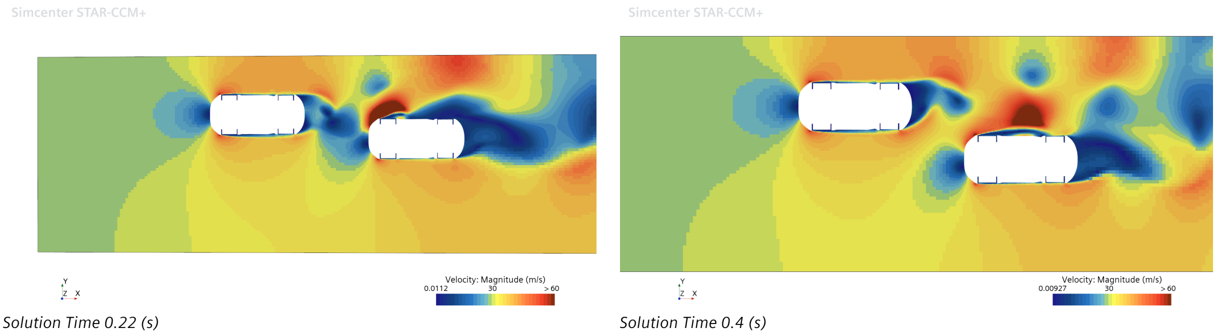
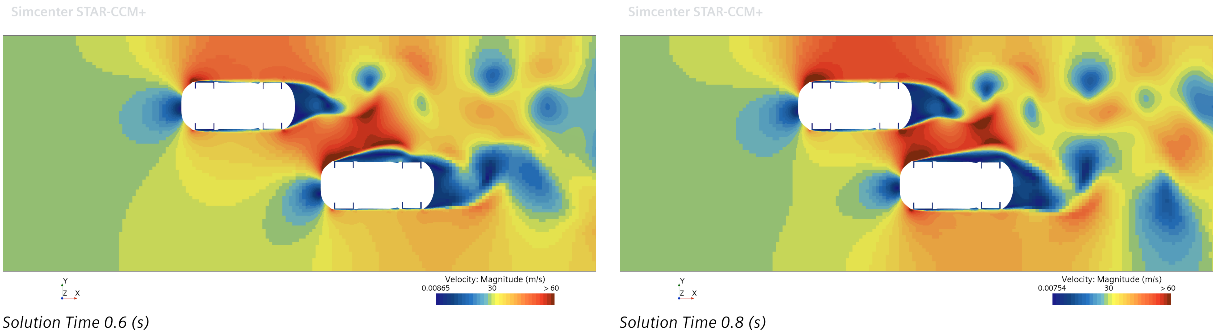


Figure 8: Static pressure field. Left: $t = 0$ s (range -3780 to $+447$ Pa) — Car 2 fully in the wake. Right: $t = 0.22$ s (range -4730 to $+761$ Pa) — low-pressure Venturi corridor generates lateral suction.

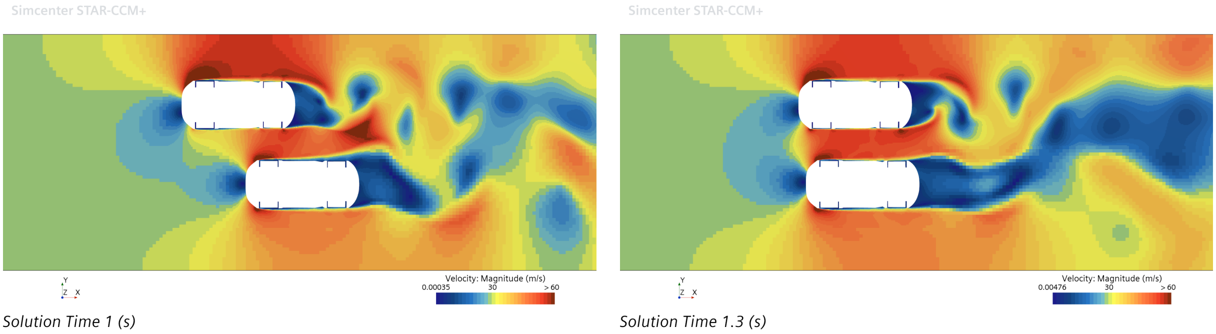
Figure 9 shows the velocity magnitude at six representative instants. Three aerodynamically distinct regimes are visible: (i) lateral emergence with gap flow acceleration; (ii) Venturi acceleration exceeding 60 m/s at maximum lateral proximity; and (iii) progressive wake decoupling as both vehicles re-establish independent far-field topologies. This progression is consistent with Watkins and Vino [2008] and Corin et al. [2008].



$t = 0.22$ s (P1) $t = 0.4$ s (P2)



$t = 0.6$ s (P3) $t = 0.8$ s (P4)



$t = 1.0$ s (P5) $t = 1.3$ s (P6)

Figure 9: Velocity magnitude (0 to > 60 m/s) at six representative instants. **P1**: lateral emergence, gap flow accelerates. **P2**: gap channel fully active, peak > 60 m/s. **P3**: maximum inboard–outboard asymmetry and peak lateral loading. **P4**: Car 2 nose past Car 1 nose. **P5**: wakes begin to decouple. **P6**: Car 2 reaches Car 1.

5.5 Drag Coefficient History

Time-averaged C_D values over six non-overlapping windows (reference velocity $U_1 = 27.78$ m/s for Car 1, $U_2 = 33.33$ m/s for Car 2; reference width $A_{\text{ref}} = 0.694$ m) are presented in Figure 10 and Table 6.

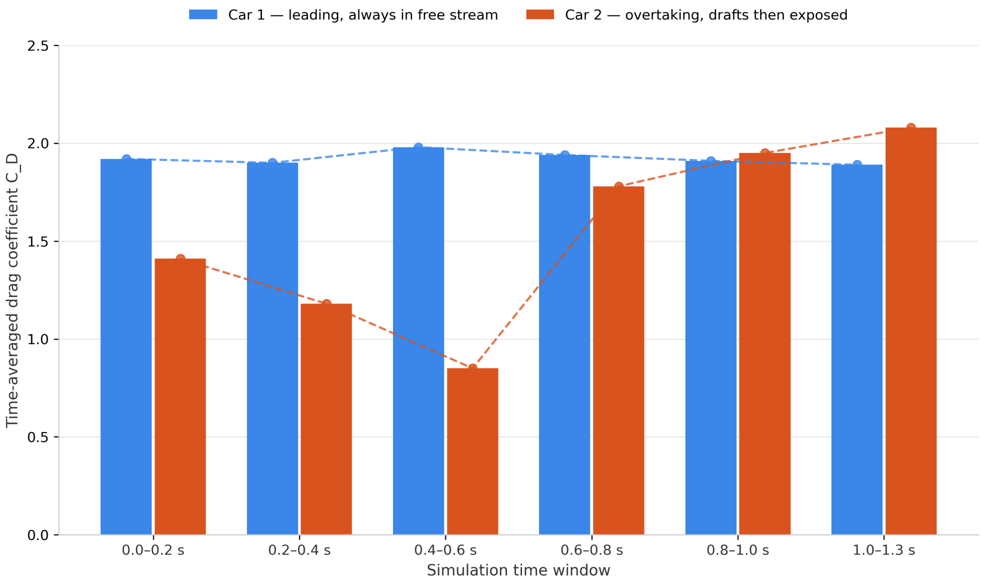


Figure 10: Time-averaged $\overline{C}_D$ for Car 1 (blue) and Car 2 (orange). Three stages: slipstreaming (0–0.6 s), transition (0.6–0.8 s), and post-overtaking stabilisation (0.8–1.3 s).

Table 6: Time-averaged drag coefficients $\overline{C}_D$.

Time window (s)	$\overline{C}_D$ **Car 1**	$\overline{C}_D$ **Car 2**
0.0 – 0.2	1.90	1.40
0.2 – 0.4	1.88	1.17
0.4 – 0.6	1.97	0.85
0.6 – 0.8	1.93	1.75
0.8 – 1.0	1.89	1.94
1.0 – 1.3	1.87	2.08

Three physically distinct stages emerge. **(1) Slipstreaming (0–0.6 s):** $\overline{C}_{D,2}$ decreases from 1.40 to 0.85 (39% reduction) as Car 2 closes on the rear of Car 1; Car 1 remains broadly stable at $\approx$ 1.90–1.97, with a slight increase at t = 0.4–0.6 s **(2) Transition (0.6–0.8 s):** $\overline{C}_{D,2}$ rises sharply from 0.85 to 1.75 (+106%) as Car 2 exits the shielding zone and its front face is exposed to the full stagnation pressure; Car 1 experiences a minor drag reduction (1.97 $\rightarrow$ 1.93). **(3) Post-overtaking stabilisation (0.8–1.3 s):** $\overline{C}_{D,2}$ continues rising to 2.08. The asymptotic value substantially exceeds 3D experimental references ($C_D \approx$ 0.30–0.40) because the 2D rectangular cross-section cannot relieve pressure drag over the roof or underbody (piston effect); cross-methodological comparison must therefore focus on *relative* trends, not absolute magnitudes. These three stages reproduce the canonical overtaking force signature of Watkins and Vino [2008] and Corin et al. [2008].

6 Concluding Remarks

6.1 Conclusions

The aerodynamic interaction during an overtaking manoeuvre has been characterised using two complementary CFD strategies. The primary conclusions are:

- **3D Quasi-Steady Force Signature:** Three aerodynamic phases were identified. In the approach phase (P1), both vehicles exhibit isolated-like values ($C_D \approx 0.35$). During lateral displacement (P2–P4), bluff-body gap blockage drives $C_{D,2}$ to a maximum of 0.437 at P4 (+24%) and $C_{D,1}$ to 0.426 at P3 (+23%); full lateral alignment (P3) is the most critical configuration for both vehicles simultaneously. The GCI study confirms discretisation uncertainty below 2.5%.
- **2D Transient Evolution:** The overset simulation reproduced the canonical three-stage force signature: a 39% drag reduction during slipstreaming (0–0.6 s), a +106% spike during the transition phase (0.6–0.8 s), and progressive post-overtaking stabilisation (0.8–1.3 s).
- **Qualitative Validity vs. Quantitative Divergence:** A significant quantitative mismatch exists between absolute magnitudes (0.35–0.44 in 3D vs. 1.87–2.08 in 2D) due to the 2D planar confinement (piston effect). The 2D model is nonetheless a qualitatively valid engineering tool; evaluations must rely on normalised relative trends.
- **Dominant Aerodynamic Mechanisms:** Three physical phenomena were isolated: wake shielding (reduces trailing-vehicle drag), bluff-body gap blockage (raises stagnation pressure), and Venturi-driven gap acceleration ($p \approx -4730$ Pa, responsible for lateral attractive suction forces).

6.2 Future Work

- **Fully Coupled 3D Transient Simulations** via HPC to eliminate the quasi-steady assumption and recover 3D vortical structures.
- **Turbulence Model Improvement:** micro-fillets at the tyre–ground contact to enable stable k-ω SST convergence, reducing modelling error from ∼21% to ∼9%.
- **Parametric Velocity Study:** quantify bow-wave intensity as a function of relative speed ΔU.
- **Realistic Geometries:** replace simplified bluff bodies with the DrivAer model [Heft et al., 2012] and add a crosswind component [Howell et al., 2012].

References

Greitzer, E. M., Tan, C. S., and Graf, M. B. (2007). *Internal Flow: Concepts and Applications*. Cambridge University Press, Cambridge.

Shih, T.-H., Liou, W. W., Shabbir, A., Yang, Z., and Zhu, J. (1995). A new k-ε eddy viscosity model for high Reynolds number turbulent flows. *Computers & Fluids*, 24(3):227–238.

Abdul-Rahman, H., Moria, H., and Rasani, M. R. (2021). Aerodynamic study of three cars in tandem using computational fluid dynamics. *Journal of Mechanical Engineering and Sciences*, 15(3):8228–8240.

GrabCAD Community (2024). Volkswagen Beetle 2016 – Replica. 3D CAD Model Library. `https://grabcad.com/library/volkswagen-beetle-2016-replica-1`. Accessed: 2024.

Browand, F., McArthur, J., and Radovich, C. (2004). Fuel saving achieved in the field test of two tandem trucks. *University of Southern California Report*, USC-CE-04-01.

Celik, I. B., Ghia, U., Roache, P. J., and Freitas, C. J. (2008). Procedure for estimation and reporting of uncertainty due to discretization in CFD applications. *Journal of Fluids Engineering*, 130(7):078001.

Corin, R. J., He, L., and Dominy, R. G. (2008). A CFD investigation into the transient aerodynamic forces on overtaking road vehicle models. *Journal of Wind Engineering and Industrial Aerodynamics*, 96(8–9):1390–1411.

Guilmineau, E. (2008). Computational study of flow around a simplified car body. *Journal of Wind Engineering and Industrial Aerodynamics*, 96(6–7):1207–1217.

Heft, A. I., Indinger, T., and Adams, N. A. (2012). Introduction of a new realistic generic car model for aerodynamic investigations. *SAE Technical Paper*, 2012-01-0168.

Howell, J., Sherwood, B., and Windsor, S. (2012). The aerodynamic interaction of a following vehicle with a leading truck. *SAE Technical Paper*, 2012-01-0171.

Hucho, W.-H. (1998). *Aerodynamics of Road Vehicles*, 4th edition. Society of Automotive Engineers, Warrendale, PA.

Menter, F. R. (1994). Two-equation eddy-viscosity turbulence models for engineering applications. *AIAA Journal*, 32(8):1598–1605.

Noger, C., Regardin, C., and Széchényi, E. (2005). Investigation of the transient aerodynamic phenomena associated with passing manoeuvres. *Journal of Fluids and Structures*, 21(3):231–241.

Schütz, T., Ed. (2015). *Aerodynamics of Road Vehicles: From Fluid Mechanics to Vehicle Engineering*, 5th edition. SAE International.

Katz, J. (2006). *Race Car Aerodynamics: Designing for Speed.* Bentley Publishers, Cambridge, MA.

Oberkampf, W. L. and Roy, C. J. (2010). *Verification and Validation in Scientific Computing.* Cambridge University Press, Cambridge.

Schito, P. and Braghin, F. (2016). Numerical and experimental investigation of trucks aerodynamics. *SAE International Journal of Commercial Vehicles*, 5(1):56–65.

Steger, J. L., Dougherty, F. C., and Benek, J. A. (1983). A chimera grid scheme. *Advances in Grid Generation, ASME FED*, 5:59–69.

Watkins, S. and Vino, G. (2008). The effect of vehicle spacing on the aerodynamics of a representative car shape. *Journal of Wind Engineering and Industrial Aerodynamics*, 96(6–7):1232–1239.

Blocken, B. and Gualtieri, C. (2012). Ten iterative steps for model development and evaluation applied to computational fluid dynamics for environmental fluid mechanics. *Environmental Modelling & Software*, 33:1–22.

A Annex I: Convergence and Sanity Checks

This annex presents the convergence evidence for all simulated cases. For each case, the evolution of the governing equation residuals and the asymptotic behaviour of C_D are shown. The 2D transient subsection also includes the steady-state initial condition and the Courant number distribution.

Note on Turbulence Model Selection

Figure 11 shows the residual history for the baseline case (P3) with the k-ω SST model. Diverging oscillations that do not stabilise within 1500 iterations confirm failure to converge, consistent with the known sensitivity of the k-ω formulation to near-wall mesh quality and free-stream ω boundary values [Menter, 1994]. The Realizable k-ε model was therefore adopted for all production runs.

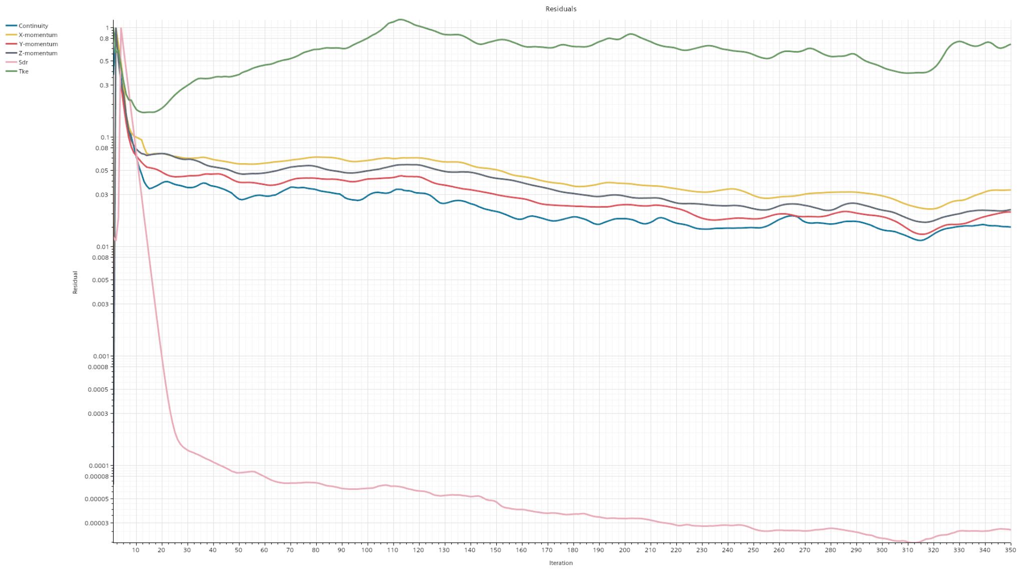


Figure 11: Residual history for the k-ω SST model on P3. Diverging oscillations indicate failure to converge.

A.1 3D Quasi-Steady Cases (Positions P1–P5)

Figure 12 and Figure 13 show the residual history and C_D monitor for position P1, taken as representative of all five quasi-steady cases. The remaining positions (P2–P5) exhibit qualitatively identical convergence behaviour and are not shown individually for brevity.

All residuals drop rapidly during the first ~400 iterations and subsequently stabilise into a low-amplitude periodic oscillation. The continuity and x-momentum equations level off below 5×10^{-4}; the turbulence quantities (Tke, Tdr) and z-momentum plateau in the range 10^{-3}–2×10^{-3}, remaining above the strict 10^{-4} threshold. This residual floor is physically consistent with the borderline quasi-steady regime ($\beta \approx 0.10$): the inter-vehicle pressure field sustains a low-frequency unsteadiness that the steady RANS solver cannot fully suppress, as discussed in Section 2.5. The C_D monitors (Figure 13) reach a stable oscillation of amplitude $\sigma < 0.001$ (Table 4), confirming that the mean values reported are statistically well-converged despite the residual oscillations.

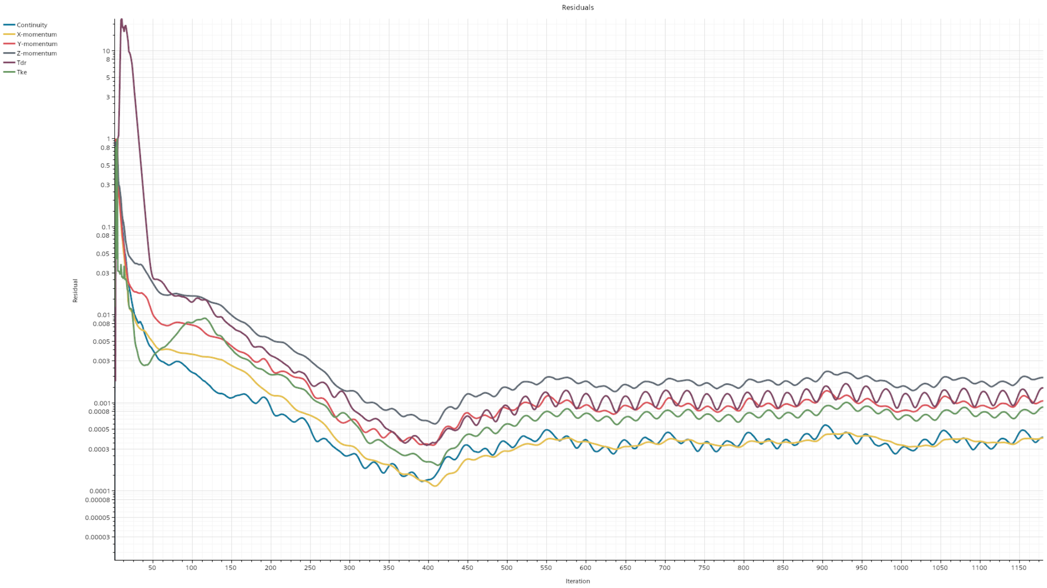


Figure 12: Residual history for position P1 (representative of P1–P5). All equations drop below 10^{-2} by iteration 200 and stabilise into a low-amplitude periodic oscillation consistent with the borderline quasi-steady regime ($\beta \approx 0.10$).

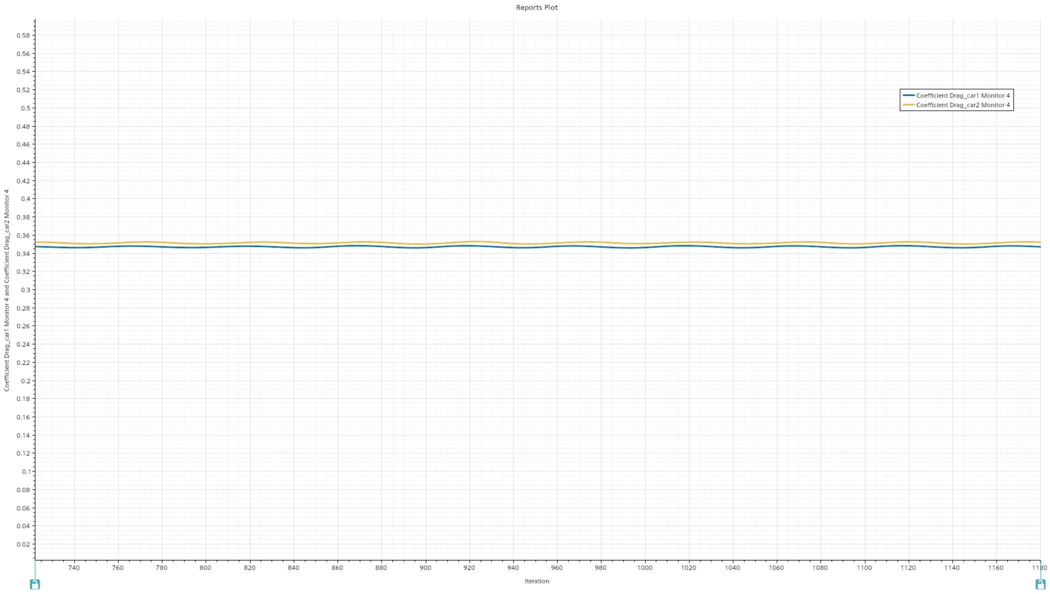


Figure 13: C_D monitor for Car 1 (blue) and Car 2 (gold) at position P1. Both signals stabilise around iteration 700 with oscillation amplitude $\sigma < 0.001$; mean values are reported in Table 4.

A.2 2D Transient Overset Case

Steady-State Initial Condition

A steady-state solution at the P1 configuration was computed prior to activating the Implicit Unsteady solver to provide a physically consistent initial condition and suppress the initial numerical transient [Corin et al., 2008]. Figure 14 confirms that Car 2 is fully immersed in the low-pressure, low-momentum wake of Car 1, reproducing the expected slipstreaming mechanism.

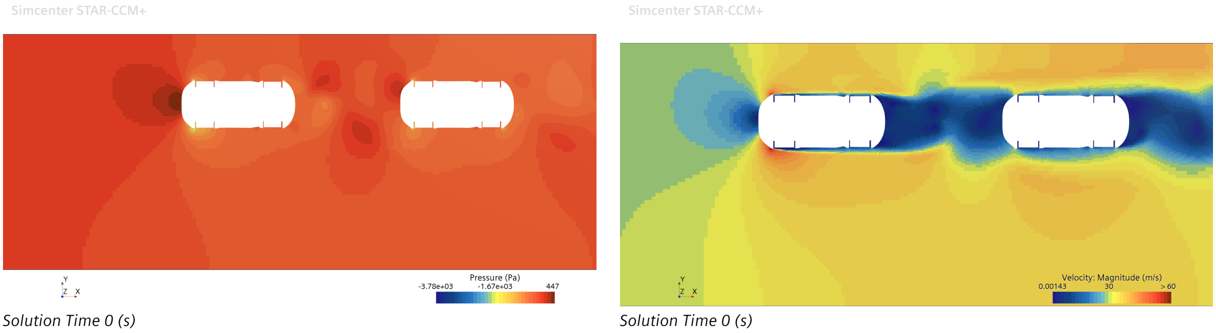


Figure 14: Steady-state initial condition at $t = 0$ s. Left: static pressure (range -3780 to $+447$ Pa); reduced stagnation pressure at Car 2's nose confirms wake shielding. Right: velocity magnitude; the low-momentum wake of Car 1 fully envelops Car 2.

Courant Number Distribution

Figure 15 shows the convective Courant number at $t = 0.226$ s. The domain is predominantly sub-unity, confirming that $\Delta t = 1 \times 10^{-4}$ s is appropriate for the bulk flow. Peak values $\mathrm{Co}_{\max} \approx 16.8$ are confined to two or three prism-layer cells at the sharp 90° rear corners of the vehicle planform (geometric singularities); the Implicit Unsteady scheme tolerates these localised values provided inner-loop residuals drop below 10^{-4} per time step [Celik et al., 2008], which was verified throughout the run.

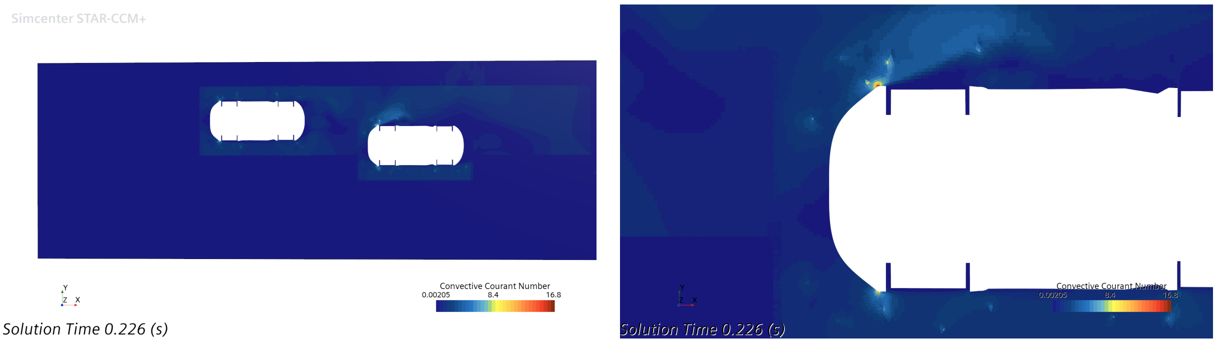


Figure 15: Convective Courant number at $t = 0.226$ s (scale 0.002–16.8). Left: global view — domain predominantly sub-unity. Right: zoomed rear corner of Car 1 where $\mathrm{Co} \approx 16.8$ is localised to two or three cells at the geometric singularity.

Residual and Force Monitor History

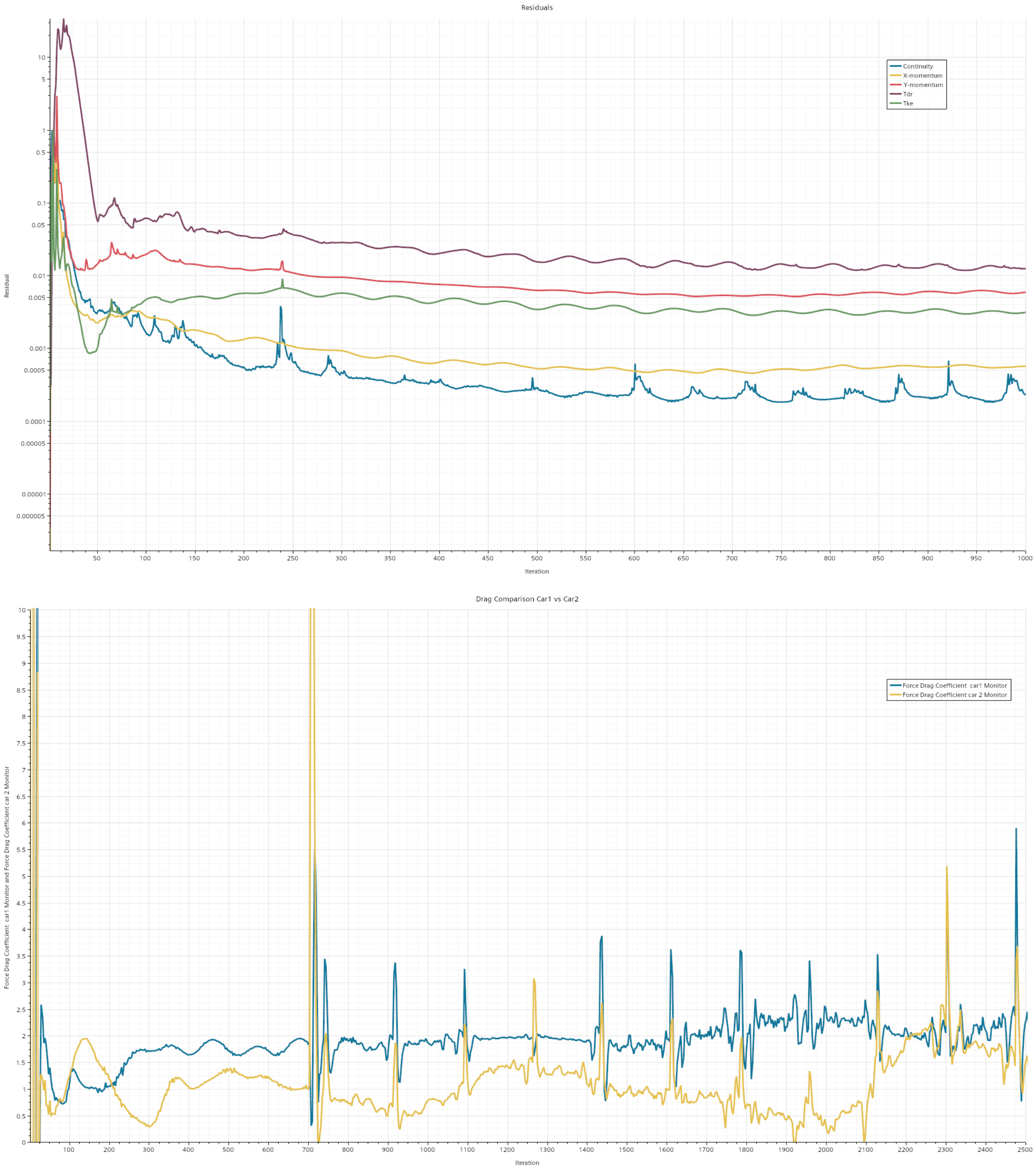


Figure 16: Above convergence of the 2D steady overset simulation. Residuals drop below 1×10^{-4} within each time step; C_D monitors show smooth temporal evolution consistent with the expected physics. Below, drag coefficient history for Car 1 and Car 2, showing the effect of their interaction over the simulation. Car 2 experiences lower drag during drafting and higher drag when exposed, while Car 1 remains comparatively more stable as the leading vehicle